

Unbinned Unfolding for Neutrino–Nucleus Cross Sections

Joseph Bailey*
Stanford Physics Department
 (Dated: March 21, 2026)

Neutrino–nucleus cross-section measurements are limited by detector smearing and model dependence inherent in the standard unfolding procedure, iterative Bayesian unfolding (IBU). We review the conceptual foundations and limitations of IBU in the context of modern neutrino experiments and examine recent developments in unbinned machine learning–based unfolding, particularly OmniFold. By replacing response matrices with multidimensional likelihood-ratio learning, unbinned approaches offer a framework for reducing prior bias and preserving correlations central to oscillation physics.

I. INTRODUCTION

Modern neutrino oscillation experiments rely on precise measurements of neutrino–nucleus cross sections. Experiments such as NOvA, T2K, and the upcoming DUNE experiment extract neutrino mixing parameters and measure possible CP violation from reconstructed energy spectra. Because these spectra depend directly on the underlying interaction cross sections, uncertainties in cross section measurements propagate into oscillation parameters and have become a source of systematic uncertainty.

At neutrino energies of a few GeV, neutrino interactions occur within complex nuclear environments. Nuclear effects—including multinucleon correlations, resonance production, and final-state interactions—modify the observable kinematics of the outgoing particles (See Fig I). These processes create strong correlations between measured quantities such as muon momentum, scattering angle, and hadronic energy, making accurate reconstruction of the underlying interaction particularly challenging.

Neutrino experiments have repeatedly shown that un-

certainities in nuclear modeling constitute one of the dominant systematic uncertainties in cross-section measurements. For example, recent MINERvA analyses demonstrate that generator assumptions enter directly into the extraction procedure through efficiency corrections, background subtraction, and unfolding itself [2]. Because these corrections rely on simulated detector responses, the statistical method used to perform unfolding plays a central role in determining the final cross section.

Neutrino experiments must therefore solve an inverse problem: the detector does not observe the true particle-level interaction directly. Instead, detectors measure reconstructed observables that have been distorted by resolution effects, limited acceptance, and detection inefficiencies. The statistical procedure used to infer the underlying distribution from these distorted measurements is known as unfolding.

Most neutrino cross-section measurements use iterative Bayesian unfolding (IBU), also known as the D’Agostini method. In this approach, simulated events are used to build a response matrix that describes how true physical quantities map to reconstructed observables. The unfolding procedure iteratively updates an estimate of the true distribution using Bayes’ theorem. While IBU is computationally practical and widely used, its reliance on finite binning and simulation-derived response matrices introduces structural limitations.

Recent developments in machine learning have proposed an alternative strategy known as unbinned unfolding. Instead of discretizing the detector response into bins, these methods learn the relationship between simulation and data directly in a high-dimensional continuous space. Among these approaches, the OmniFold framework has attracted significant attention in collider physics because it uses neural networks to estimate likelihood ratios between simulated and observed data distributions.

This Letter reviews the conceptual foundations of traditional unfolding in neutrino experiments and examines the emerging paradigm of unbinned machine learning–based unfolding. We focus on how these approaches differ in their treatment of dimensionality, prior dependence, and nuclear-model systematics, and we discuss the implications for precision neutrino oscillation experiments.

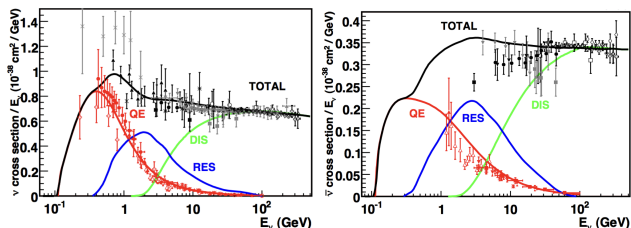


FIG. 1. Contributions to the neutrino–nucleus cross section as a function of neutrino energy, illustrating the coexistence of quasielastic (QE), resonance (RES), and deep inelastic scattering (DIS) processes. The presence of multiple interaction mechanisms produces complex correlations among final-state observables. From [1]

* josephrb@stanford.edu

II. ITERATIVE BAYESIAN UNFOLDING AND ITS STRUCTURAL CONSTRAINTS

Because detectors observe only the final-state particles produced after complex nuclear interactions and detector effects, the underlying particle-level distributions must be inferred statistically. In matrix-based unfolding, the detector response is represented by a response matrix R_{ij} defined as

$$R_{ij} \approx P(x_i^{\text{reco}} | x_j^{\text{true}}),$$

the probability that an event originating in true bin j is reconstructed in bin i . The observed reconstructed counts are therefore modeled as

$$N_i^{\text{reco}} = \sum_j R_{ij} N_j^{\text{true}} + B_i,$$

where B_i denotes the estimated number of background events in reconstructed bin i . The unfolding problem consists of estimating the unknown true distribution N_j^{true} from the measured counts N_i^{reco} .

Direct matrix inversion is generally unstable because the response matrix is typically ill-conditioned and may include inefficiencies or bin migrations. Instead, most neutrino cross-section measurements employ iterative Bayesian unfolding (IBU), also known as the D'Agostini method. In this approach, Bayes' theorem is used to invert the detector response probabilistically. Given the forward detector mapping $P(x_i^{\text{reco}} | x_j^{\text{true}})$, Bayes' theorem gives

$$P(x_j^{\text{true}} | x_i^{\text{reco}}) = \frac{P(x_i^{\text{reco}} | x_j^{\text{true}}) P(x_j^{\text{true}})}{\sum_k P(x_i^{\text{reco}} | x_k^{\text{true}}) P(x_k^{\text{true}})}.$$

Substituting the response matrix yields

$$P(x_j^{\text{true}} | x_i^{\text{reco}}) = \frac{R_{ij} \pi_j}{\sum_k R_{ik} \pi_k},$$

where $\pi_j = P(x_j^{\text{true}})$ is the prior distribution for the true events, typically obtained from a neutrino event generator.

Using this conditional probability, the unfolded event counts can be written as

$$\hat{N}_j = \sum_i U_{ji} (N_i^{\text{reco}} - B_i),$$

where the effective unfolding operator

$$U_{ji} \approx P(x_j^{\text{true}} | x_i^{\text{reco}})$$

is determined by the Bayesian update. In general $U_{ji} \neq R_{ij}^{-1}$; instead, it represents a probabilistic inversion of the detector response. The procedure is performed iteratively by updating the prior after each step,

$$\pi_j^{(t+1)} = \frac{\hat{N}_j^{(t)}}{\sum_k \hat{N}_k^{(t)}},$$

allowing the estimate of the true distribution to gradually approach a stable solution. The number of iterations therefore acts as a regularization parameter: too few iterations retain bias toward the initial prior, while too many amplify statistical fluctuations.

The unfolded counts are converted into a differential cross section through

$$\frac{d\sigma}{dx_j} = \frac{\hat{N}_j}{\Phi_\nu T \Delta x_j \epsilon_j},$$

where Φ_ν is the integrated neutrino flux, T is the number of target nuclei, Δx_j is the bin width, and ϵ_j is the selection efficiency.

In principle the detector response is a continuous conditional probability density $P(\vec{x}_{\text{reco}} | \vec{x}_{\text{true}})$. In practice this mapping is discretized into the finite matrix R_{ij} defined over a limited set of observables. While this discretization makes the inverse problem computationally tractable, it introduces several structural limitations.

First, binning reduces dimensionality. Neutrino interactions produce correlated kinematic structures involving muon momentum, scattering angle, hadronic energy, and particle multiplicities. When unfolding is performed in a single observable, the remaining degrees of freedom are effectively marginalized, and even two-dimensional unfoldings cannot fully capture these correlations.

Second, finite bin widths introduce discretization bias. Narrow bins improve resolution but require large simulated samples to maintain statistical stability of the response matrix. Limiting the number of iterations suppresses statistical fluctuations but increases bias toward the initial prior, producing the familiar bias-variance trade-off.

Third, prior dependence is intrinsic to the method. Because the initial distribution is derived from a neutrino event generator, differences in nuclear modeling can propagate into the unfolded result, particularly when the unfolding is strongly regularized.

These limitations reflect the difficulty of representing a high-dimensional stochastic detector response with a finite response matrix. As experimental precision improves, this motivates approaches that operate directly on the underlying probability densities without explicit binning.

III. UNBINNED LIKELIHOOD-RATIO LEARNING AND THE OMNIFOLD FRAMEWORK

The OmniFold method approaches unfolding from a different perspective by avoiding explicit discretization of the detector response. Rather than explicitly constructing a discretized response matrix, OmniFold frames unfolding as a problem of density-ratio estimation between experimental data and simulated events.

A machine-learning classifier is trained to distinguish reconstructed experimental events from reconstructed

simulated events. In doing so the classifier learns a function proportional to the likelihood ratio

$$r(\vec{x}) = \frac{p_{\text{data}}(\vec{x})}{p_{\text{sim}}(\vec{x})},$$

where $\vec{x}$ denotes the reconstructed observables. This ratio determines how simulated events must be reweighted so that their reconstructed distribution matches the observed data.

The resulting weights are applied to simulated events such that

$$p_{\text{sim}}(\vec{x}) \rightarrow p_{\text{data}}(\vec{x}),$$

and are then propagated to the corresponding particle-level events. Because detector effects are stochastic, the procedure alternates between detector-level reweighting and particle-level reweighting in an iterative loop.

Conceptually, this converts unfolding into a forward-modeling problem: the particle-level distribution associated with the iteratively reweighted simulation is interpreted as the unfolded estimate of the true distribution. Because the method only requires evaluating the detector simulation in the forward direction, it avoids explicitly inverting the detector response—eliminating the ill-conditioned matrix inversion that constrains traditional approaches.

Unlike matrix-based unfolding, this approach operates directly on continuous observables and therefore avoids discretizing the detector response. The neural network effectively replaces the response matrix with a learned likelihood ratio in high-dimensional feature space, allowing multidimensional correlations among observables to be incorporated naturally. Regularization arises from the iteration procedure and training dynamics rather than from binning choices.

By replacing the discretized response matrix with likelihood-ratio learning in continuous feature space, unbinned methods such as OmniFold provide a framework capable of preserving multidimensional correlations that are difficult to capture within traditional matrix-based approaches.

IV. IMPLICATIONS FOR NEUTRINO PHYSICS

The distinction between IBU and unbinned likelihood-ratio methods is not merely technical, but structural. In IBU, detector smearing is explicitly encoded in a finite matrix defined over selected observables. Conversely, in OmniFold, the detector mapping is learned implicitly through the requirement that the reweighted simulation matches data in a high-dimensional continuous space.

Consequently, the sources of bias and the mechanisms of regularization differ fundamentally. While IBU bias arises from finite binning, strict prior dependence, and iteration truncation, OmniFold's bias stems from limited

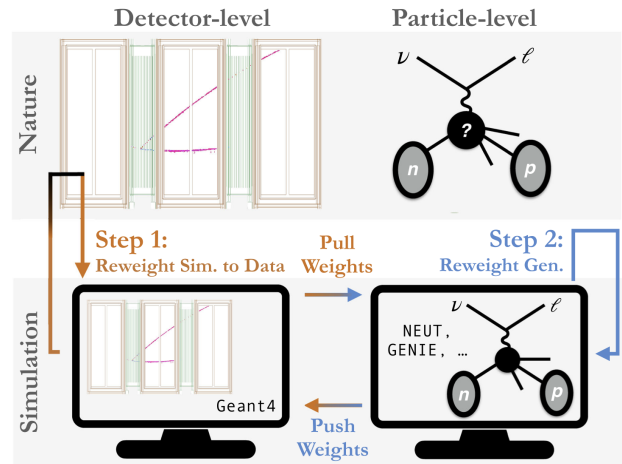


FIG. 2. The OmniFold unfolding procedure. Simulated particle-level events are passed through a detector simulation to produce reconstructed observables that can be directly compared with experimental data. A classifier is trained to distinguish reconstructed data from reconstructed simulation, allowing the network to learn the likelihood ratio $r(\vec{x}) = p_{\text{data}}(\vec{x})/p_{\text{sim}}(\vec{x})$. The resulting event weights are applied to the corresponding particle-level simulated events. The procedure alternates between detector-level reweighting and particle-level reweighting until the simulated reconstructed distribution matches the observed data. The unfolded particle-level distribution is then obtained from the final weighted simulation sample. From [3].

network capacity, training instabilities, or imperfect optimization.

For neutrino-nucleus scattering, the preservation of multidimensional correlations is uniquely relevant. Complex processes like multinucleon interactions and final-state interactions couple lepton and hadron kinematics in nontrivial ways. If unfolding is restricted to a single lepton variable—as is often practically required in IBU—hadronic information is marginalized and integrated out. This potentially embeds theoretical generator assumptions directly into the effective response. An unbinned method that simultaneously incorporates multiple correlated observables may better capture these kinematic structures without artificial data compression.

As a preliminary demonstration, we applied OmniFold with three iterations to MINERvA charged-current inclusive data, unfolding the muon transverse momentum p_T^μ . When compared against the standard MINERvA IBU result and the generator-level truth distribution, the unbinned method produces a ratio to truth consistent with unity across the measured range, performing comparably to IBU in this single-variable case. While a one-dimensional unfolding does not yet exercise the multidimensional advantages of the method, it establishes baseline consistency and motivates the extension to higher-dimensional phase spaces where correlations between lep-

tonic and hadronic observables become important.

At the same time, machine learning-based unfolding introduces distinct new considerations. Neural networks require careful validation to avoid overfitting, and statistical uncertainties must be robustly propagated through the entire training procedure. Interpretability and reproducibility remain central concerns in precision cross-section measurements. The absence of explicit binning does not remove the need for rigorous systematic evaluation of underlying modeling assumptions.

Thus, unbinned methods should not currently be viewed as drop-in replacements for traditional approaches without thorough validation in a neutrino-specific context. Rather, they represent a conceptual extension that naturally aligns with the high dimensionality of modern detector data and complex nuclear modeling.

V. DISCUSSION AND OUTLOOK

Future long-baseline neutrino experiments aim to measure CP violation and the neutrino mass ordering with unprecedented precision. Achieving these goals requires cross section measurements with correspondingly small systematic uncertainties. As experimental precision improves, biases introduced during the unfolding stage will increasingly limit the accuracy of oscillation measurements.

As discussed in Sections II and III, the structural limitations of matrix-based unfolding—finite binning, prior

dependence, and dimensional compression—become increasingly consequential as experimental precision improves.

Despite these advantages, machine learning-based unfolding introduces new challenges. Neural networks must be carefully validated to ensure that they do not simply reproduce the assumptions embedded in the simulated training samples. Robust uncertainty propagation, statistical stability, and reproducibility remain critical requirements for precision measurements.

For neutrino physics, the next step is systematic benchmarking of unbinned methods against established approaches. Closure tests, generator stress tests, and comparisons across different nuclear interaction models will be essential for quantifying potential biases and evaluating uncertainty propagation. Such studies will determine whether the additional flexibility of machine learning-based unfolding leads to genuinely improved cross-section measurements.

The broader methodological shift from low-dimensional matrix inversion to high-dimensional likelihood-ratio learning reflects a growing convergence between particle physics and modern statistical inference. If validated within neutrino experiments, unbinned unfolding may provide a powerful framework for extracting physics from increasingly complex detector data in the precision oscillation era.

ACKNOWLEDGMENTS

Used ChatGPT on the first draft to sound more like “Physical Review Letter”.

-
- [1] J. L. Hewett, H. Weerts, R. Brock, J. N. Butler, B. C. K. Casey, J. Collar, A. de Gouvea, R. Essig, Y. Grossman, W. Haxton, J. A. Jaros, C. K. Jung, Z. T. Lu, K. Pitts, Z. Ligeti, J. R. Patterson, M. Ramsey-Musolf, J. L. Ritchie, A. Roodman, K. Scholberg, C. E. M. Wagner, G. P. Zeller, S. Aefsky, A. Afanasev, K. Agashe, C. Albright, J. Alonso, C. Ankenbrandt, M. Aoki, C. A. Argüelles, N. Arkani-Hamed, J. R. Armendariz, C. Armendariz-Picon, E. A. Diaz, J. Asaadi, D. M. Asner, K. S. Babu, K. Bailey, O. Baker, B. Balantekin, B. Baller, M. Bass, B. Batell, J. Beacham, J. Behr, N. Berger, M. Bergevin, E. Berman, R. Bernstein, A. J. Bevan, M. Bishai, M. Blanke, S. Blessing, A. Blondel, T. Blum, G. Bock, A. Bodek, G. Bonvicini, F. Bossi, J. Boyce, R. Breedon, M. Breidenbach, S. J. Brice, R. A. Briere, S. Brodsky, C. Bromberg, A. Bross, T. E. Browder, D. A. Bryman, M. Buckley, R. Burnstein, E. Cadén, P. Campana, R. Carlini, G. Carosi, C. Castromonte, R. Cenci, I. Chakaberia, M. C. Chen, C. H. Cheng, B. Choudhary, N. H. Christ, E. Christensen, M. E. Christy, T. E. Chupp, E. Church, D. B. Cline, T. E. Coan, P. Coloma, J. Comfort, L. Coney, J. Cooper, R. J. Cooper, R. Cowan, D. F. Cowen, D. Cronin-Hennessy, A. Datta, G. S. Davies, M. Demarteau, D. P. DeMille, A. Denig, R. Dermisek, A. Deshpande, M. S. Dewey, R. Dharma-
- palan, J. Dhooghe, M. R. Dietrich, M. Diwan, Z. Djurcic, S. Dobbs, M. Duraisamy, B. Dutta, H. Duyang, D. A. Dwyer, M. Eads, B. Echenard, S. R. Elliott, C. Escobar, J. Fajans, S. Farooq, C. Faroughy, J. E. Fast, B. Feinberg, J. Felde, G. Feldman, P. Fierlinger, P. F. Perez, B. Filippone, P. Fisher, B. T. Flemming, K. T. Flood, R. Forty, M. J. Frank, A. Freyberger, A. Friedland, R. Gandhi, K. S. Ganezer, A. Garcia, F. G. Garcia, S. Gardner, L. Garrison, A. Gasparian, S. Geer, V. M. Gehman, T. Gershon, M. Gilchriese, C. Ginsberg, I. Gogoladze, M. Gonderinger, M. Goodman, H. Gould, M. Graham, P. W. Graham, R. Gran, J. Grange, G. Gratta, J. P. Green, H. Greenlee, R. C. Group, E. Guardincerri, V. Gudkov, R. Guenette, A. Haas, A. Hahn, T. Han, T. Handler, J. C. Hardy, R. Harnik, D. A. Harris, F. A. Harris, P. G. Harris, J. Hartnett, B. He, B. R. Heckel, K. M. Heeger, S. Henderson, D. Hertzog, R. Hill, E. A. Hinds, D. G. Hitlin, R. J. Holt, N. Holtkamp, G. Horton-Smith, P. Huber, W. Huelsnitz, J. Imber, I. Irastorza, J. Jaeckel, I. Jaegle, C. James, A. Jawahery, D. Jensen, C. P. Jessop, B. Jones, H. Jostlein, T. Junk, A. L. Kagan, M. Kalita, Y. Kamyshev, D. M. Kaplan, G. Karagiorgi, A. Karle, T. Katori, B. Kayser, R. Kephart, S. Kettell, Y. K. Kim, M. Kirby, K. Kirch, J. Klein, J. Kneller, A. Kobach, M. Kohl, J. Kopp, M. Kordosky, W. Korsch, I. Kourba-

- nis, A. D. Krisch, P. Krizan, A. S. Kronfeld, S. Kulkarni, K. S. Kumar, Y. Kuno, T. Kutter, T. Lachenmaier, M. Lamm, J. Lancaster, M. Lancaster, C. Lane, K. Lang, P. Langacker, S. Lazarevic, T. Le, K. Lee, K. T. Lesko, Y. Li, M. Lindgren, A. Lindner, J. Link, D. Lissauer, L. S. Littenberg, B. Littlejohn, C. Y. Liu, W. Loinaz, W. Lorenzon, W. C. Louis, J. Lozier, L. Ludovici, L. Lueking, C. Lunardini, D. B. MacFarlane, P. A. N. Machado, P. B. Mackenzie, J. Maloney, W. J. Marciano, W. Marsh, M. Marshak, J. W. Martin, C. Mauger, K. S. McFarland, C. McGrew, G. McLaughlin, D. McKeen, R. McKeown, B. T. Meadows, R. Mehdiev, D. Melconian, H. Merkel, M. Messier, J. P. Miller, G. Mills, U. K. Minamisono, S. R. Mishra, I. Mocioiu, S. M. Sher, R. N. Mohapatra, B. Monreal, C. D. Moore, J. G. Morfin, J. Mousseau, L. A. Moustakas, G. Mueller, P. Mueller, M. Muether, H. P. Mumm, C. Munger, H. Murayama, P. Nath, O. Naviliat-Cuncin, J. K. Nelson, D. Neuffer, J. S. Nico, A. Norman, D. Nygren, Y. Obayashi, T. P. O'Connor, Y. Okada, J. Olsen, L. Orozco, J. L. Orrell, J. Osta, B. Pahlka, J. Paley, V. Papadimitriou, M. Papucci, S. Parke, R. H. Parker, Z. Parsa, K. Partyka, A. Patch, J. C. Pati, R. B. Patterson, Z. Pavlovic, G. Paz, G. N. Perdue, D. Perevalov, G. Perez, R. Petti, W. Pettus, A. Piepke, M. Pivovarov, R. Plunkett, C. C. Polly, M. Pospelov, R. Povey, A. Prakesh, M. V. Purohit, S. Raby, J. L. Raaf, R. Rajendran, S. Rajendran, G. Rameika, R. Ramsey, A. Rashed, B. N. Ratcliff, B. Rebel, J. Redondo, P. Reimer, D. Reitzner, F. Ringer, A. Ringwald, S. Riordan, B. L. Roberts, D. A. Roberts, R. Robertson, F. Robicheaux, M. Rominsky, R. Roser, J. L. Rosner, C. Rott, P. Rubin, N. Saito, M. Sanchez, S. Sarkar, H. Schellman, B. Schmidt, M. Schmitt, D. W. Schmitz, J. Schneps, A. Schopper, P. Schuster, A. J. Schwartz, M. Schwarz, J. Seeman, Y. K. Semertzidis, K. K. Seth, Q. Shafi, P. Shanahan, R. Sharma, S. R. Sharpe, M. Shiozawa, V. Shiltsev, K. Sigurdson, P. Sikivie, J. Singh, D. Sivers, T. Skwarnicki, N. Smith, J. Sobczyk, H. Sobel, M. Soderberg, Y. H. Song, A. Soni, P. Souder, A. Sousa, J. Spitz, M. Stancari, G. C. Stavenga, J. H. Steffen, S. Stepanyan, D. Stoeckinger, S. Stone, J. Strait, M. Strassler, I. A. Sulai, R. Sundrum, R. Svoboda, B. Szczerbinska, A. Szec, T. Takeuchi, P. Tanedo, S. Taneja, J. Tang, D. B. Tanner, R. Tayloe, I. Taylor, J. Thomas, C. Thorn, X. Tian, B. G. Tice, M. Tobar, N. Tolich, N. Toro, I. S. Towner, Y. Tsai, R. Tschirhart, C. D. Tunnell, M. Tzanov, A. Upadhye, J. Urheim, S. Vahsen, A. Vainshtein, E. Valencia, R. G. V. de Water, R. S. V. de Water, M. Velasco, J. Vogel, P. Vogel, W. Vogelsang, Y. W. Wah, D. Walker, N. Weiner, A. Weltman, R. Wendell, W. Wester, M. Weinstein, C. White, L. Whitehead, J. Whitmore, E. Widmann, G. Wiedemann, J. Wilkerson, G. Wilkinson, P. Wilson, R. J. Wilson, W. Winter, M. B. Wise, J. Wodin, S. Wojcicki, B. Wojtsekhowski, T. Wongjirad, E. Worcester, J. Wurtele, T. Xin, J. Xu, T. Yamanaka, Y. Yamazaki, I. Yavin, J. Yeck, M. Yeh, M. Yokoyama, J. Yoo, A. Young, E. Zimmerman, K. Zioutas, M. Zisman, J. Zupan, and R. Zwaska, *Fundamental physics at the intensity frontier* (2012), arXiv:1205.2671 [hep-ex].
- [2] L. Zazueta, S. Akhter, Z. Ahmad Dar, F. Akbar, V. Ansari, M. V. Ascencio, M. S. Athar, A. Bashyal, A. Bercellie, M. Betancourt, J. L. Bonilla, A. Bravar, T. Cai, G. A. Díaz, H. da Motta, J. Felix, L. Fields, A. Filkins, R. Fine, A. M. Gago, H. Gallagher, A. Ghosh, R. Gran, E. Granados, D. A. Harris, S. Henry, D. Jena, S. Jena, J. Kleykamp, A. Klustová, M. Kordosky, D. Last, A. Lozano, X.-G. Lu, E. Maher, S. Manly, W. A. Mann, K. S. McFarland, B. Messerly, J. Miller, O. Moreno, J. G. Morfin, J. K. Nelson, C. Nguyen, A. Olivier, V. Paolone, G. N. Perdue, K.-J. Plows, M. A. Ramírez, D. Ruterbories, H. Schellman, C. J. Solano Salinas, H. Su, M. Sultana, E. Valencia, N. H. Vaughan, A. V. Waldron, and B. Yaeggy (MINER ν A Collaboration), Improved constraint on the MINER ν A medium energy neutrino flux using $\bar{\nu}e^- \rightarrow \bar{\nu}e^-$ data, *Phys. Rev. D* **107**, 012001 (2023).
- [3] R. G. Huang, A. Cudd, M. Kawaue, T. Kikawa, B. Nachman, V. Mikuni, and C. Wilkinson, Machine learning assisted unfolding for neutrino cross-section measurements with the omnifold technique, *Phys. Rev. D* **112**, 012008 (2025).